Large Language Models for Aspect-Based Sentiment Analysis of Uzbek Service Sector Reviews

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Abstract. Online consumer reviews are a primary form of digitally mediated social-behavioural data, yet for low-resource languages such as Uzbek the signal is largely inaccessible due to the absence of fine-grained sentiment resources. We present **UzABSA-LLM**, an end-to-end framework that fine-tunes open-source LLMs with QLoRA for Aspect-Based Sentiment Analysis (ABSA) on Uzbek service-sector reviews and turns the strongest model into a scalable multi-domain annotation engine. Under an identical QLoRA configuration we compare Qwen-2.5-7B, Llama-3.1-8B and DeepSeek-R1-Distill-Qwen-7B against their own zero-/few-shot baselines on a 618-sentence test split; fine-tuned Qwen reaches ATE-F1 0.660 with a 100% JSON parse rate, while Llama reaches the best aspect–polarity F1 0.581 and sentiment accuracy 0.886. Strikingly, validation cross-entropy ranking does not match ABSA task ranking. We then deploy the strongest model over 5,038 multi-domain reviews (23 business categories), score 307 of them with a GPT-4o-mini LLM-as-Judge five-dimensional rubric, and human-verify 50 reviews (94% gold-agreement, Spearman ρ=0.71 judge–human). The resulting silver corpus of 9,884 aspects surfaces sector-specific satisfaction patterns in banking, retail and HoReCa directly interpretable as social signals. We release the dataset, code and adapters.

Keywords: Aspect-Based Sentiment Analysis · Large Language Models · Uzbek · Low-resource NLP · QLoRA · LLM-as-Judge · Social-behavioural data.

1 Introduction

The rapid expansion of digital platforms has transformed online consumer reviews into a primary source of social-behavioural data. Citizens use these platforms to evaluate services, articulate satisfaction or dissatisfaction, and communicate expectations toward businesses and public institutions. In emerging digital societies, such reviews can support evidence-based service improvement and offer a window into how trust and quality of experience are constructed in everyday interactions [9,28].

For low-resource languages, however, this signal is largely inaccessible. Aspect-Based Sentiment Analysis (ABSA) [19], the task of jointly identifying *what* is being evaluated (an *aspect*) and *how* it is evaluated (its *polarity*), is the natural computational lens for this kind of analysis because it preserves fine-grained social information that flat positive/negative classification discards. Yet ABSA research is dominated by English and a handful of high-resource languages [9], and Uzbek – a Turkic, agglutinative language with a dual Latin/Cyrillic literacy and an emerging digital ecosystem – remains a striking blind spot. Multilingual encoders such as mBERT and XLM-R offer only marginal Uzbek representation [6,2]; domain-adapted Uzbek BERTs exist [12,15] but are general-purpose and have not been applied to token-level ABSA. The first dedicated Uzbek ABSA dataset, UzABSA, was released only recently [18] and covers a single domain (restaurant reviews).

Two recent technical developments make it timely to bridge this gap. First, open-weight LLMs at the 7–8B parameter scale (Qwen-2.5 [20], Llama-3.1 [7], DeepSeek-R1 distillations [4]) show emergent cross-lingual generalisation, and instruction-tuned ABSA [23,26] reformulates ABSA as constrained text generation, side-stepping the need for tag-set engineering. Second, parameter-efficient fine-tuning through QLoRA [5] reduces a 7B fine-tuning job to < 6 GB of GPU memory, which makes systematic comparison feasible even under low-resource computational budgets.

We bring these two threads together for Uzbek and frame the result as *computational social-behavioural research.* Our contributions are the following.

1. **First systematic LLM-based ABSA study for Uzbek.** We fine-tune and compare three open-source LLMs (Qwen-2.5-7B, Llama-3.1-8B, DeepSeek-R1-Distill-Qwen-7B) under an identical QLoRA configuration on the UzABSA SemEval-2014-style corpus, and benchmark them against their own zero-shot and 3-shot baselines.
2. **A scalable model-assisted annotation pipeline.** We chain the best fine-tuned model (Qwen-2.5-7B, 100% JSON parse rate) with an LLM-as-Judge quality layer (GPT-4o-mini, five-dimensional rubric) [27,13] and a human verification sample, producing a quality-tiered silver standard at a marginal cost.
3. **A novel multi-domain silver Uzbek ABSA dataset.** 5,038 reviews spanning 23 business categories (banking, retail, HoReCa, telecom, healthcare, education, e-commerce, transport, ...) are released with 9,884 aspect–sentiment pairs, per-review judge scores, and provenance metadata.
4. **Empirical and social-behavioural findings.** (i) Training cross-entropy ranking does not match ABSA task ranking, which has implications for model selection in structured extraction. (ii) A restaurant-trained model generalises to 15 of 23 service domains above an LLM-judged inclusion threshold, with quality degrading along a measurable in-domain $\rightarrow$ near-domain $\rightarrow$ specialised-domain gradient. (iii) Aspect-level satisfaction patterns are sector-specific and directly interpretable as actionable evidence for service redesign.

2 Related Work

ABSA: from sequence labelling to instruction tuning. ABSA was formalised by the SemEval-2014 Task 4 shared task [19], which decomposed the problem into Aspect Term Extraction (ATE), Aspect Category Detection (ACD), Aspect Sentiment Classification (ASC) and end-to-end joint extraction. Subsequent work explored sequence-to-sequence formulations [14], graph-convolution models [25], few-shot category detection [8] and BERT-based joint models [3]. Recent surveys [9,28] report that more than 80% of ABSA studies use English data, with the restaurant and laptop domains dominating. InstructABSA [23] re-framed ABSA as instruction learning, demonstrating that a single instruction-tuned model can solve multiple ABSA subtasks competitively. We adopt the same generation-based formulation but at a larger parameter scale, in a non-English low-resource setting, and target a single end-to-end joint output.

Uzbek NLP and Uzbek ABSA. Resources for Uzbek have grown steadily [10,16,17,24,1], including sentiment datasets [11], a Latin/Cyrillic transliteration tool [22], an evaluation benchmark [21], and Uzbek-specific encoders such as BERTbek [12] and TahrirchiBERT [15]. UzABSA, the first SemEval-style Uzbek ABSA dataset, was released by [18] but contains only restaurant reviews and has not yet been used to train a generative LLM. Our work both consumes UzABSA as a training resource and extends it to 23 business domains through model-assisted annotation.

QLoRA, LLM-as-Judge, and service-review data. QLoRA [5] combines 4-bit NF4 quantisation with Low-Rank Adapters, enabling 7B fine-tuning on a single GPU; LLM surveys [26] confirm that small, high-quality instruction datasets suffice for structured extraction. Using a strong LLM to score weaker outputs (LLM-as-Judge) was popularised by MT-Bench and Chatbot Arena [27] but is subject to position and verbosity biases [13], requiring human calibration. We follow this practice, pairing GPT-4o-mini judgements with a 50-review human verification step. Finally, online reviews are digitally-mediated behavioural traces [9] whose aspect-level disaggregation provides the granularity at which businesses and policy-makers operate.

3 Data: A Social-Behavioural Corpus of Uzbek Service Reviews

The data backbone of this work has two complementary parts, summarised in Table 1. The first is the manually annotated, single-domain UzABSA dataset [18], which provides the supervision signal for fine-tuning. The second is a multi-domain raw corpus collected from the Uzbek consumer-review platform `sharh.commeta.uz`, which provides the breadth needed for cross-domain evaluation and for the multi-domain silver dataset.

UzABSA (annotated). The publicly released dataset [18] contains 6,175 Uzbek review sentences in SemEval-2014 Task 4 format with character-level aspect-term spans, four polarity labels (*positive*, *negative*, *neutral*, *conflict*) and five aspect categories (*ovqat*, *muhit*, *xizmat*, *narx*, *boshqalar*). The positive class dominates ($\approx 64\%$), so we report per-class as well as micro-averaged metrics. We use a deterministic 80/10/10 split with `seed=42`.

Multi-domain raw corpus. The raw corpus is collected from `sharh.commeta.uz`. After de-duplication and a 10-character minimum, 5,038 reviews from 630 businesses remain. A keyword taxonomy maps each business to one of 23 domains (Table 2), with the three target sectors well represented: HoReCa (Restoran/Ovqatlanish 1,626, Sayohat 78), banking (Bank/Moliya 577, To'lov tizimlari 328), retail (Elektron tijorat 296, Oziq-ovqat 215, Gul/Sovg'a 173). Script distribution is 92.78% Uzbek-Latin, 5.18% Russian-Cyrillic, 2.04% Uzbek-Cyrillic, motivating the tokenizer-fertility analysis in Section 6.

Domain-aware aspect taxonomy. Because aspects are intrinsically domain-bound (*kredit* matters in banking but not in HoReCa; *ovqat_ sifati* matters in HoReCa but not in telecom), we build a hierarchical taxonomy of 119 sub-aspects across 18 domains (released alongside the dataset). For training we keep the original 5-category UzABSA taxonomy to remain compatible with [18]; for qualitative interpretation in Section 7 we map the predicted categories back into the domain-aware taxonomy.

Ethics and licensing. All reviews are publicly accessible from the source platform; only the textual content and business name are retained. Personal identifiers are not collected, and reviewer names from the source are not redistributed in the silver dataset.

4 Method

4.1 ABSA as instruction-tuned generation

We model joint ABSA as constrained text generation, in the spirit of InstructABSA [23] but at the 7–8B parameter scale. Given an input sentence in Uzbek, the model must produce a JSON object listing all aspects with their category and polarity. Listing 1.1 shows the production prompt (Uzbek system instruction, ChatML template) and a target output. JSON is chosen for machine-parseability: parse failure is itself an instruction-following signal that we report.

4.2 QLoRA fine-tuning configuration

All three models are fine-tuned under an *identical* configuration so that the only varied factor across runs is the base model. The base weights are loaded in 4-bit NormalFloat (NF4) with double quantisation; gradients backpropagate through

Listing 1.1. Production prompt (abridged) and target output. The system message is in Uzbek; the assistant message is a strict JSON object.

```
<|im_start|>system
Siz o'zbek tilida matnlardan aspektlarni va ularning hissiyotlarini
aniqlash bo'yicha mutaxassissiz. Javobni qat'iy JSON formatida bering.
<|im_end|>
<|im_start|>user
Quyidagi matndan aspektlarni, kategoriyalarni va hissiyot polaritesini
aniqlang. Matn: "Bu restoranning ovqatlari juda mazali, lekin narxlari
qimmat."
<|im_end|>
<|im_start|>assistant
{"aspects": [
  {"term": "ovqatlari", "category": "ovqat", "polarity": "positive"},
  {"term": "narxlari", "category": "narx",  "polarity": "negative"}
]}
<|im_end|>
```

the quantised weights into LoRA adapters with rank $r = 16$, $\alpha = 32$, dropout 0.05 and zero bias, attached to all attention and FFN projections (`q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj`). Compute precision is BF16; optimisation uses 8-bit AdamW, cosine schedule, peak LR 2×10^{-4}, warmup ratio 0.1, weight decay 0.01, gradient clipping 1.0, packing disabled. Effective batch size is 16 (per-device 4 $\times$ gradient accumulation 4), max sequence length 2,048 tokens, 1,000 optimisation steps (approximately 3 epochs over the 80% training split), seed 42. We use the Unsloth [20] fused-kernel implementation on top of HuggingFace TRL.

4.3 Three base models

We select three open-weight, instruction-tuned 7–8B models that span distinct pre-training stories and tokenisers, and that all admit 4-bit loading: **Qwen-2.5-7B-Instruct** [20] (strong multilingual coverage), **Llama-3.1-8B-Instruct** [7] (Western-language flagship) and **DeepSeek-R1-Distill-Qwen-7B** [4] (a reasoning-distilled variant sharing the Qwen-7B backbone). The DeepSeek variant lets us isolate the contribution of the reasoning distillation relative to the same architecture under instruction tuning.

4.4 Three-layer model-assisted annotation pipeline

Once the strongest fine-tuned model is identified (Section 6.3), we re-use it as an *annotator* for the 5,038-review multi-domain corpus. Three quality layers stack on top of each other:

- **Layer 1 – ABSA generation.** Greedy decoding ($T = 0.1$, `do_sample=False`, `max_new_tokens`=512) over every review; outputs parsed as JSON, then post-filtered to drop empty-aspect predictions and flag reviews with > 5 aspects.

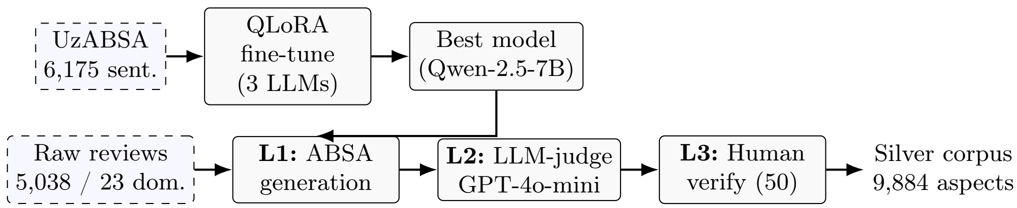

Fig. 1. End-to-end UzABSA-LLM pipeline. Three LLMs are QLoRA-fine-tuned on UzABSA; the best (Qwen-2.5-7B) is re-deployed as an annotator for the multi-domain raw corpus, gated by an LLM-as-Judge layer and calibrated by a 50-review human verification step.

- **Layer 2 – LLM-as-Judge.** On a stratified 307-review sample (domain-balanced; at least three reviews per domain), GPT-4o-mini scores each prediction on a five-dimensional 1–5 rubric: *Completeness*, *Accuracy*, *Sentiment correctness*, *Relevance* of the category, and an *Overall* score. The judge sees the source text, business name and rating, but *not* a gold reference – this is the real-world setting [27], and the rubric is designed to be aspect-agnostic. We adopt the include/flag/exclude tiers at ≥ 3.5 / $[2.5, 3.5)$ / < 2.5 on the Overall dimension.
- **Layer 3 – Human calibration.** A 50-review subset (stratified across domains and judge tiers) is independently labelled by the authors as `correct`/`incorrect` on the predicted aspects. Spearman ρ is computed between the judge Overall score and the binary human verdict.

The output of the pipeline is a quality-tiered silver dataset published in JSON and JSONL formats with full provenance (annotator model id, judge score, human-verified flag). Figure 1 summarises the three layers end-to-end.

5 Experimental Setup

Hardware and software. All fine-tuning and inference runs use a single NVIDIA RTX A6000 (48 GB VRAM, Ampere SM 8.6) under CUDA 12.8. The stack is PyTorch 2.x, Transformers $\geq$ 4.45, TRL (SFTTrainer), PEFT, bitsandbytes, Unsloth. Training experiments are logged to Weights & Biases; Figure 2 shows the side-by-side W&B dashboard used during the campaign.

Metrics. For ATE we report micro-averaged exact-match and partial-match Precision/Recall/F1. For Aspect Sentiment Classification (ASC) on matched aspect pairs we report accuracy and macro-F1. For end-to-end ABSA we report the joint *aspect–polarity pair* F1 (E2E-F1), our primary metric. For TASD (Target-Aspect-Sentiment Detection) we report micro-F1 over (term, category, polarity) triples, restricted to the $\approx 75\%$ subset of sentences where positional alignment between the term and category lists is unambiguous. We always report the *JSON parse success rate* as a first-class metric: an unparseable output simply

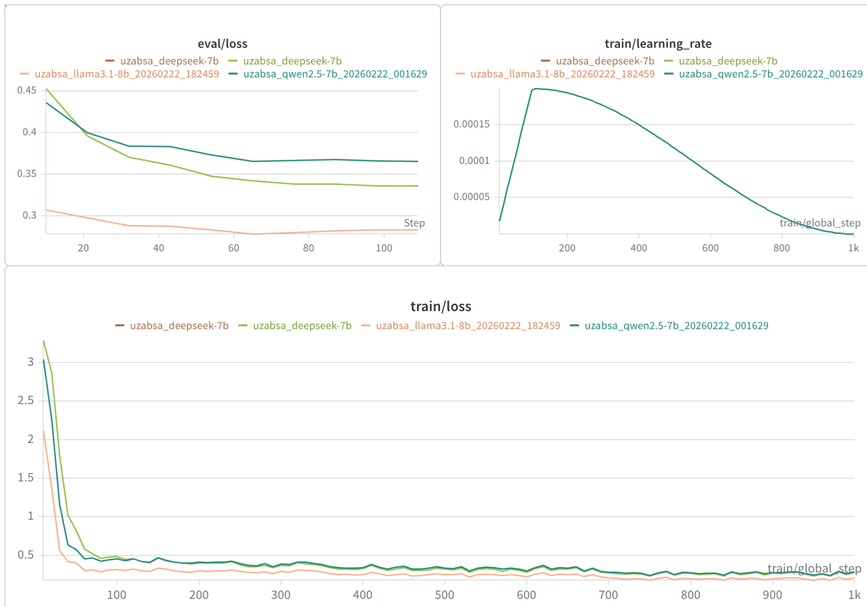

Fig. 2. Weights & Biases dashboard for the three QLoRA fine-tuning runs. Each model is trained for 1,000 steps under an identical configuration; only the base model differs. The dashboard tracks training loss, evaluation loss, learning-rate schedule, gradient norm and GPU memory in real time, supporting the controlled comparison reported in Section 6.3.

cannot enter downstream analysis and must therefore be visible in the headline table. Greedy decoding is used at evaluation time. Statistical comparisons are bootstrapped over 1,000 resamples of the test set.

Baselines. For each base model we run a zero-shot and a 3-shot baseline on the test split using the same Uzbek instruction prompt as in fine-tuning; the 3-shot exemplars are sampled from train with `seed=42` and kept fixed across all three models.

Judge configuration. The judge is `gpt-4o-mini` (temperature 0.0, deterministic decoding). Each judgement is a single API call returning structured JSON. We re-prompt up to twice on parse failure; remaining failures are excluded from aggregate statistics (0 of 307 in our final run).

6 Results

6.1 Tokenizer fertility on Uzbek

We measure tokenisation efficiency by the standard fertility metric – tokens per whitespace-delimited word – on both corpora and across the three script buckets identified in Section 3. As Table 3 shows, the two Qwen-family tokenisers are roughly 17% more compact than Llama-3.1 on Uzbek-Latin and up to 24% more compact on Uzbek-Cyrillic. Fertility correlates with the zero-shot baseline ATE-F1 (Section 6.2): Llama under-performs Qwen at zero-shot despite a higher parameter count. Importantly, however, the gap narrows substantially after fine-tuning (Section 6.3), showing that tokenizer disadvantage is partially compensated by gradient-based adaptation – a finding consistent with cross-lingual transfer results [2].

6.2 Zero-/few-shot baselines vs. fine-tuned

Table 4 reports zero-shot and 3-shot results for the three base models against their QLoRA-fine-tuned counterparts. Three observations stand out. (i) *Fine-tuning is the dominant factor*: the smallest fine-tuning gain over 3-shot is +0.30 ATE-F1 (Llama). (ii) *Few-shot prompting helps modestly* (+0.05 to +0.09 ATE-F1 over zero-shot), but does not close the gap with fine-tuning. (iii) *Instruction-following fails out of the box*: zero-shot JSON parse rates are between 41% and 72%, vs. 95–100% after fine-tuning. For real-world annotation, this difference alone justifies the fine-tuning cost.

6.3 Fine-tuned model comparison

Table 5 reports the controlled three-model comparison on the full test set with matched training-efficiency figures. Qwen-2.5-7B is the strongest aspect extractor (ATE-F1 0.660 exact, 0.770 partial) and produces syntactically valid JSON on *every* test sentence. Llama-3.1-8B edges out Qwen on the joint aspect–polarity F1 (0.5805 vs. 0.5795) and on sentiment classification (accuracy 0.886, macro-F1 0.844) but fails to produce parseable JSON on 4.1% of inputs. DeepSeek-R1-Distill underperforms its sibling Qwen-2.5 on every ABSA metric despite *better* validation cross-entropy than Qwen-2.5 – a clear empirical illustration that loss-based model selection can mislead for structured-output tasks. All three runs sit within an 8.5 GB GPU envelope (well under the A6000's 48 GB), so the choice is driven by task metrics rather than compute.

Loss-versus-task dissociation. Llama owns the best validation loss (0.279) but *not* the best ATE-F1; DeepSeek owns better validation loss than Qwen (0.336 vs. 0.366) but loses across all ABSA metrics, in part because it over-predicts (predicted/reference ratio 1.29 vs. 1.16 for Qwen and 1.05 for Llama). The literature has documented similar dissociations for QA and code generation [26]; we provide a clean Uzbek ABSA counterpart and recommend reporting both loss *and* downstream metrics in low-resource fine-tuning studies.

TASD on the aligned subset. Because the UzABSA aspect-term and aspect-category lists are aligned in $\approx 75\%$ of sentences (those with equal-length lists), we can recover (term, category, polarity) triplets by positional alignment and report TASD micro-F1. Llama is the strongest end-to-end predictor (P/R/F1 =

0.502/0.531/**0.516**), narrowly ahead of Qwen (0.481/0.553/0.514) and well ahead of DeepSeek (0.398/0.510/0.447), in line with the E2E Pair-F1 ranking above.

6.4 Multi-domain LLM-as-Judge quality matrix

We now move from the annotated test split to the open multi-domain corpus. The best fine-tuned model (Qwen-2.5-7B, selected for ATE recall and 100% parse rate) is run over all 5,038 reviews, producing 9,884 aspects with a polarity distribution of 75.1% positive / 16.5% negative / 8.4% neutral. A 307-review stratified sample is then judged by GPT-4o-mini. Table 6 reports the aggregate scores and tier distribution; Table 7 reports per-domain Overall scores for the sectors emphasised by this paper.

Three patterns emerge. (a) The model trained on restaurant reviews *generalises* above the 3.5 inclusion threshold for 15 out of 23 domains, including the core retail (Elektron tijorat, Yetkazib berish) and banking (Bank/Moliya, To'lov tizimlari) sectors. (b) *Accuracy* is uniformly high ($\geq 3.67/5$) across all domains: when the model extracts an aspect, the aspect genuinely appears in the text. The quality gradient is driven by *Completeness* – the model misses domain-specific aspects rather than inventing wrong ones. (c) The largest residual gap is in highly specialised verticals (insurance, investment), where the lexical distance from restaurant reviews is largest. We discuss the social-behavioural implications of this gradient in Section 7.

6.5 Human verification (Layer 3)

Table 8 shows that 47 of 50 human-checked predictions are substantively correct, and that the judge's Overall score correlates strongly with the human verdict (Spearman $\rho = 0.71$). This supports both the silver dataset's empirical quality and the use of the LLM judge as a scalable proxy for human inspection, in line with prior findings [27]. The three failures are concentrated in highly specialised verticals (one insurance, one investment, one telecom), matching the per-domain gradient of Section 6.4.

7 Discussion: A Social-Behavioural Reading

What the silver corpus says about Uzbek service experience. Aggregated over 9,884 aspects, the silver corpus contains a striking positivity bias (75.1% positive). This mirrors the survivorship bias of public review platforms (satisfied users post more reviews than dissatisfied ones), but the 16.5% negative class still encodes a clear social signal once disaggregated by sector. In **HoReCa**, the most frequent positive aspects are *ovqat* (food quality) and *xizmat* (service), and the most frequent negatives are *narx* (price) and *muhit* (ambience-related concerns about cleanliness or noise). In **banking and fintech**, positives concentrate on *xizmat_ko'rsatish* and *ilova_qulayligi* (app usability), while negatives concentrate on *tezlik* (response speed) and *kredit* (loan conditions). In **retail and delivery**, positives concentrate on *mahsulot_sifati* (product quality) and *tezlik*, and negatives on *yetkazib_berish* (delivery) and *narx*. These patterns are consistent with cross-cultural ABSA studies in other emerging markets [3] but provide the *first* Uzbek-language quantitative profile of which dimensions of service most strongly drive (dis-)satisfaction.

Loss does not equal social signal, and completeness is the bottleneck. The Llama-vs-Qwen dissociation matters socially as well as technically: a study selecting on lowest validation loss would have picked Llama, missing the perfect-parse, higher-recall behaviour of Qwen that makes the multi-domain annotation pipeline feasible at scale. The judge dimensions (Table 6) further decompose the residual error into a distinctive pattern: high accuracy and sentiment correctness, lower completeness and relevance – the model rarely fabricates concerns citizens did not raise, but under-counts those they did. Under-counting preserves precision at the cost of recall and motivates domain-adaptive re-training as the most impactful future direction.

Limitations. (i) UzABSA does not annotate opinion terms, so we cannot evaluate the ASTE/ACOS/ASQP triplet/quad subtasks; this is shared with several SemEval-2014-derived resources but is the largest single annotation gap. (ii) Our silver dataset is annotated by a model fine-tuned on a single domain; per-domain quality matters and is reported, but full gold annotation for the non-HoReCa domains remains future work. (iii) The LLM judge is itself fallible; we calibrate it against 50 human-verified reviews but acknowledge the documented biases of LLM-judge configurations [13]. (iv) The 5,058-review base corpus is modest in absolute terms; expanding it is a priority for follow-up work and is technically straightforward given the established pipeline.

8 Conclusion and Future Work

We presented UzABSA-LLM, the first systematic study of LLM-based ABSA for Uzbek. Under an identical QLoRA configuration, Qwen-2.5-7B, Llama-3.1-8B and DeepSeek-R1-Distill-Qwen-7B were trained and benchmarked against their own zero-/few-shot baselines. Fine-tuning dominates (zero-shot ATE-F1 $\approx$ 0.25 vs. fine-tuned $\approx$ 0.66), and Qwen vs. Llama trades JSON robustness against sentiment accuracy. The strongest model was re-deployed as an annotator over 5,038 multi-domain reviews, producing a quality-tiered silver corpus of 9,884 aspects across 23 domains; GPT-4o-mini judging plus a 50-review human-verification subset (47/50 correct, Spearman ρ=0.71) bound its empirical quality. We extract the first Uzbek-language aspect-level satisfaction profile for banking, retail and HoReCa and argue that validation cross-entropy is an unreliable selection criterion for structured-output ABSA.

Future work targets three directions: (i) *domain-adaptive ABSA* incorporating the 119-subcategory taxonomy via self-training on the silver dataset; (ii) *opinion-term annotation* to enable ASTE/ACOS evaluation; and (iii) *cross-lingual* *transfer* to other Turkic languages (Karakalpak, Kazakh, Kyrgyz, Turkmen) to test whether the in-domain → specialised-domain gradient observed for Uzbek replicates [2,17].

Acknowledgments. We thank the `sharh.commeta.uz` community whose public reviews made this study possible. Compute was provided by the National University of Uzbekistan.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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Table 1. Corpus statistics. The annotated split follows SemEval-2014 Task 4 format; the raw multi-domain corpus is used both for tokenizer profiling and as the substrate for the model-assisted annotation pipeline of Section 4.

Resource	**Description**	**Size**
UzABSA annotated [18]	Single-domain (restaurant) Uzbek ABSA, SemEval-2014 format	6,175 sent.
Aspect terms (positive / negative / neutral / conflict)	4,153 / 1,555 / 1,601 / 103	7,412
Aspect categories (5 labels)	ovqat, muhit, xizmat, narx, boshqalar	7,724
Train / Dev / Test split (80/10/10, seed 42)	4,940 / 617 / 618 sentences	6,175
Raw multi-domain reviews	sharh.commeta.uz, 630 businesses, 23 domains	5,058
after dedup./length filter	input to annotation pipeline	5,038
mean words / chars per review	13.55 / 96.26	—
Uzbek-Latin / Russian-Cyrillic / Uzbek-Cyrillic	92.78% / 5.18% / 2.04%	—

Table 2. Top business domains in the multi-domain corpus (5,038 reviews → 23 domains). Domains explicitly named in the conference framing – banking, retail, HoReCa – are well represented.

Domain	**Reviews**	**%**	**Domain**	**Reviews**	**%**
Restoran/Ovqatlanish (HoReCa)	1,626	32.3	Telekommunikatsiya	234	4.6
Bank/Moliya (banking)	577	11.5	Oziq-ovqat do'konlari	215	4.3
Boshqa	507	10.1	Gul/Sovg'a	173	3.4
To'lov tizimlari (fintech)	328	6.5	Texnologiya/Media	110	2.2
Elektron tijorat (retail)	296	5.9	Sayohat/Turizm	78	1.6
Ta'lim	287	5.7	Sport/Fitnes	60	1.2
Tibbiyot/Sog'liqni saqlash	285	5.7	Other 9 domains	262	5.2

Table 3. Tokenizer fertility (tokens per whitespace-separated word) on Uzbek text, computed on the UzABSA annotated set and on the 5,038-review multi-domain corpus, broken down by dominant script. Lower is better. Qwen-family tokenisers are markedly more compact on Uzbek-Latin than the Llama tokeniser, especially in the Cyrillic buckets.

Tokenizer	**Annotated**	**Uzbek-Latin**	**Russian-Cyr.**	**Uzbek-Cyr.**
Qwen-2.5-7B	1.71	1.74	1.95	2.18
DeepSeek-R1-Distill-Qwen-7B	1.72	1.75	1.96	2.19
Llama-3.1-8B	2.03	2.07	2.42	2.71

Table 4. Zero-shot, 3-shot and fine-tuned ABSA performance on the 618-sentence held-out test split. ATE-F1 is exact-match; E2E-F1 is the joint aspect–polarity pair F1; "%JSON" is the JSON parse success rate. Best per column in **bold**.

	0-shot			**3-shot**			**Fine-tuned**		
Model	ATE-F1	E2E-F1	%JSON	ATE-F1	E2E-F1	%JSON	ATE-F1	E2E-F1	%JSON
Qwen-2.5-7B	0.282	0.181	71.6	0.341	0.243	84.3	**0.660**	0.580	**100.0**
Llama-3.1-8B	0.246	0.158	56.8	0.336	0.230	78.7	0.655	**0.581**	95.9
DeepSeek-R1-Distill-7B	0.207	0.122	41.4	0.291	0.187	67.2	0.603	0.502	95.4

Table 5. Fine-tuned three-model ABSA comparison on the 618-sentence test split, with matched QLoRA training-efficiency figures. "Pair F1" is E2E aspect–polarity F1 (primary metric); "Acc."/"Mac-F1" are sentiment-classification metrics on matched pairs; "Loss" is the best evaluation cross-entropy; "Min" is wall-clock training time.

	ATE F1			**Sent.**			**Train**	
Model	Exact	Partial	**Pair F1**	Acc.	Mac-F1	%JSON	Loss	Min
Qwen-2.5-7B	**0.660**	**0.770**	0.580	0.878	0.811	**100.0**	0.366	**49.0**
Llama-3.1-8B	0.655	0.759	**0.581**	**0.886**	**0.844**	95.9	**0.279**	59.8
DeepSeek-R1-Distill-7B	0.603	0.728	0.502	0.832	0.772	95.4	0.336	62.3

Table 6. LLM-as-Judge aggregate scores (1–5 scale; 307 stratified reviews; GPT-4o-mini). Accuracy is high across the board; the bottleneck is *Completeness*, i.e. aspects that exist in the text but are missed by the model.

	Mean dimension score (1–5)					**Tier** (%)		
	Comp.	Acc.	Sent.	Rel.	Overall	Incl.	Flag	Excl.
Aggregate	3.32	**4.47**	4.21	4.06	3.75	63.5	28.0	8.5

Table 7. Per-domain LLM-as-Judge Overall scores for the three sectors foregrounded by this paper, plus selected near-domain and out-of-domain comparisons. "N" is the per-domain sample size used by the judge. Domains above the 3.5 inclusion threshold are shaded grey.

Domain	**N**	Comp.	Acc.	Sent.	Rel.	**Overall**
HoReCa (in/near-domain for Layer 1)						
Restoran/Ovqatlanish	50	3.68	4.76	4.56	4.66	**4.24**
Oziq-ovqat do'konlari	15	3.47	4.33	4.47	4.13	3.80
Retail						
Elektron tijorat	21	3.43	4.48	4.10	4.14	3.81
Gul/Sovg'a	12	3.08	4.08	4.83	3.67	3.58
Banking and finance						
Bank/Moliya	30	3.40	4.47	4.17	3.90	3.70
To'lov tizimlari	24	3.00	4.29	4.38	3.54	3.54
Other out-of-domain						
Tibbiyot/Sog'liqni saqlash	25	3.40	4.60	4.20	4.36	3.80
Telekommunikatsiya	25	3.04	4.36	3.64	3.64	3.32
Ta'lim	20	2.90	4.20	3.80	3.45	3.25

Table 8. Human verification on a 50-review subset, stratified across domains and judge tiers. Authors independently labelled each review as *correct* (aspects accurately reflect text) or *incorrect*. Spearman ρ is computed between the judge Overall score and the binary human verdict on the same 50 reviews.

Quantity	**Value**
Reviews labelled	50
Correct	47 / 50 (94.0%)
Incorrect	3 / 50 (6.0%)
Spearman ρ (judge Overall vs. human verdict)	0.71